

Normal Prefetchers vs. the Trace-Local Neural Prefetcher

All normal-prefetcher cases; best-normal vs. NN on accuracy, timeliness, coverage, IPC and their averages; and how every NN indicator moves with model size

Data: v4_9_5_one_shot_all_seed7 keyed ChampSim replays (A100, seed 7)

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1 Normal prefetchers — every case, tables and graphs

Nine baselines (**no_pref** + eight prefetchers) were replayed on all five traces under the identical 25M+25M window used for the **NN**. Tables first, then the graphs.

Table 1: Normal prefetchers on 602.gcc.s-734B — all cases, all requested metrics

prefetcher	IPC	speedup	accuracy	timeliness	event cov.	issued	useful	useless	late	L2	missrate
no_pref	0.36800	1	0	0	0	0	0	0	0		0.5025
stride	0.37006	1.0056	0.2642	0.9976	0.1716	267,553	70,694	1,023	168		0.4163
streamer	0.43161	1.1729	0.1037	0.9999	0.9234	3,623,513	375,582	16,394	49		0.0383
ampm	0.43069	1.1704	0.1311	0.9984	0.9157	2,846,050	373,090	12,600	604		0.0421
spp	0.42836	1.1640	0.0590	0.9985	0.6508	4,491,857	265,086	1,162	387		0.1753
ipcp	0.36796	0.9999	0.2447	0.9200	0	188	46	74	4		0.5025
sms	0.42213	1.1471	0.4587	0.9994	0.7971	696,706	319,593	23,409	200		0.1018
sandbox	0.43628	1.1855	0.1016	0.9998	0.9644	3,839,474	390,102	32,524	59		0.0177
power7	0.43075	1.1705	0.1128	0.9997	0.9144	3,304,558	372,751	13,449	128		0.0428

Table 2: Normal prefetchers on 605.mcf.s-994B — all cases, all requested metrics

prefetcher	IPC	speedup	accuracy	timeliness	event cov.	issued	useful	useless	late	L2	missrate
no_pref	0.18318	1	0	0	0	0	0	0	0		0.7513
stride	0.18359	1.0022	0.3023	0.9999	0.0066	25,125	7,595	167	1		0.7464
streamer	0.18350	1.0017	0.0871	0.8947	0.1159	1,775,122	154,695	686,999	18,210		0.7120
ampm	0.18874	1.0304	0.1775	0.9858	0.0779	509,840	90,495	58,586	1,305		0.6974
spp	0.18874	1.0304	0.0202	0.9196	0.0070	710,459	14,384	5,178	1,257		0.7464
ipcp	0.18318	1	0.2955	0.6190	0	88	26	16	16		0.7513
sms	0.18777	1.0251	0.1991	0.8074	0.1238	1,287,493	256,319	635,113	61,126		0.6924
sandbox	0.16905	0.9229	0.0290	0.7631	0.2132	15,501,838	449,411	7,188,334	139,527		0.7249
power7	0.18163	0.9915	0.0885	0.8796	0.0959	1,466,157	129,697	556,672	17,755		0.7266

Table 3: Normal prefetchers on 619.lbm_s-4268B — all cases, all requested metrics

prefetcher	IPC	speedup	accuracy	timeliness	event cov.	issued	useful	useless	late	L2 missrate
no_pref	0.32200	1	0	0	0	0	0	0	0	0.9987
stride	0.32465	1.0082	0.4304	0.9057	0.2282	350,381	150,814	0	15,702	0.7702
streamer	0.36368	1.1294	0.7656	0.7811	0.4825	524,800	401,794	990	112,603	0.5163
ampm	0.37480	1.1640	0.2471	0.9516	0.7991	2,008,610	496,363	348	25,272	0.1989
spp	0.37933	1.1780	0.0501	0.7451	0.2332	3,804,639	190,662	10	65,229	0.7655
ipcp	0.32200	1	0.0864	0.8750	0	81	7	3	1	0.9987
sms	0.38105	1.1834	0.9421	0.9535	0.8664	568,017	535,130	84	26,099	0.1312
sandbox	0.37440	1.1627	0.1456	0.8632	0.7094	3,352,602	488,051	1,296	77,378	0.2891
power7	0.36128	1.1220	0.5992	0.8274	0.5441	615,642	368,891	862	76,926	0.4545

Table 4: Normal prefetchers on 620.omnetpp_s-874B — all cases, all requested metrics

prefetcher	IPC	speedup	accuracy	timeliness	event cov.	issued	useful	useless	late	L2 missrate
no_pref	0.23750	1	0	0	0	0	0	0	0	0.7013
stride	0.23833	1.0035	0.3028	1	0.0094	19,579	5,928	1,849	0	0.6949
streamer	0.24228	1.0201	0.1463	0.9776	0.1624	757,318	110,831	286,074	2,538	0.6165
ampm	0.23976	1.0095	0.1454	0.9603	0.0691	310,419	45,147	98,338	1,868	0.6648
spp	0.23869	1.0050	0.0004	0.6852	0.0002	174,696	74	53	34	0.7008
ipcp	0.23771	1.0009	0.1852	0.9349	0.0068	29,016	5,374	9,433	374	0.6981
sms	0.24695	1.0398	0.2447	0.9819	0.2717	808,086	197,737	394,038	3,655	0.5456
sandbox	0.24433	1.0288	0.0490	0.9718	0.4235	7,623,344	373,815	3,279,950	10,848	0.5265
power7	0.24232	1.0203	0.1444	0.9746	0.1621	730,714	105,531	271,622	2,751	0.6180

Table 5: Normal prefetchers on 623.xalancbmk_s-700B — all cases, all requested metrics

prefetcher	IPC	speedup	accuracy	timeliness	event cov.	issued	useful	useless	late	L2 missrate
no_pref	0.35321	1	0	0	0	0	0	0	0	0.3696
stride	0.35340	1.0005	0.0629	0.7154	0.0047	299,845	18,849	373	7,498	0.3679
streamer	0.33644	0.9525	0.0452	0.9426	0.0569	2,459,972	111,140	715,938	6,770	0.5994
ampm	0.34924	0.9888	0.0572	0.9417	0.0418	924,845	52,884	86,456	3,276	0.4151
spp	0.35391	1.0020	0.0047	0.8069	0.0025	1,670,201	7,816	23,108	1,870	0.3608
ipcp	0.35241	0.9977	0.0673	0.9803	0.0063	87,172	5,863	10,856	118	0.3785
sms	0.34167	0.9673	0.1509	0.9634	0.1523	1,932,085	291,589	764,626	11,070	0.5387
sandbox	0.34524	0.9774	0.0416	0.9579	0.3658	12,891,224	536,632	4,491,270	23,596	0.4924
power7	0.34330	0.9719	0.0466	0.9169	0.0383	1,764,449	82,274	477,415	7,460	0.5051

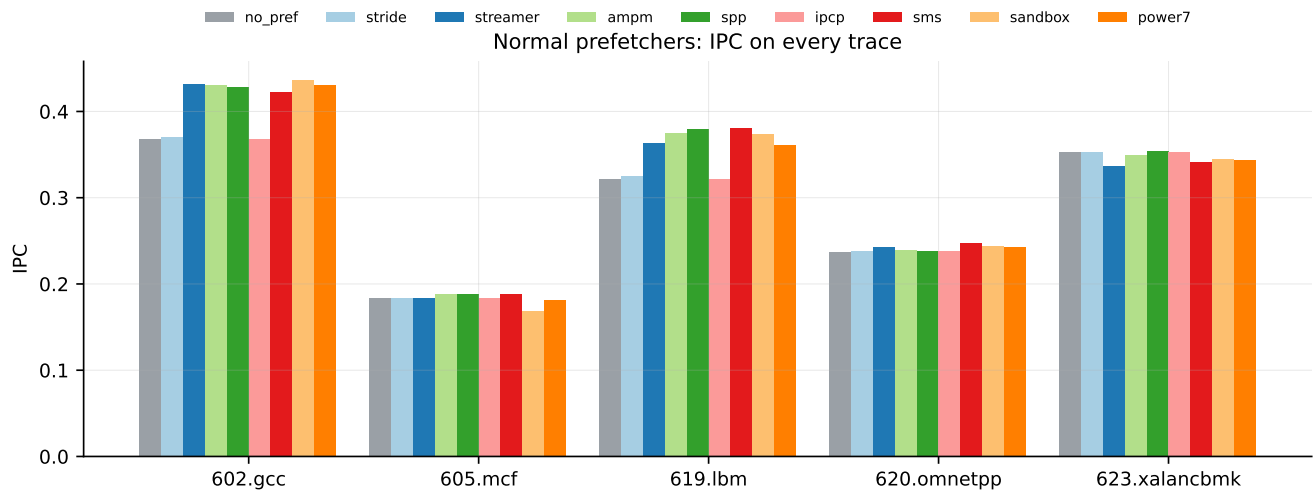


Figure 1: IPC of every normal prefetcher on every trace. Different traces crown different champions: sandbox (602), AMPM (605), SMS (619, 620), SPP (623) — no single normal wins everywhere.

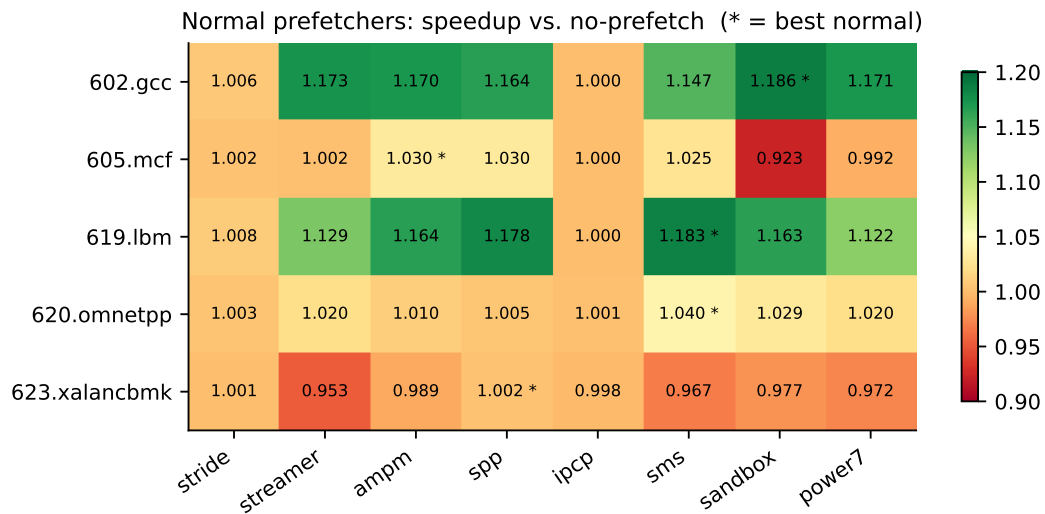


Figure 2: Speedup vs. no-prefetch for all 40 (trace, prefetcher) cases; the per-trace best normal is starred. Note the red cells: on 623 five of eight normals are *slower than not prefetching*.

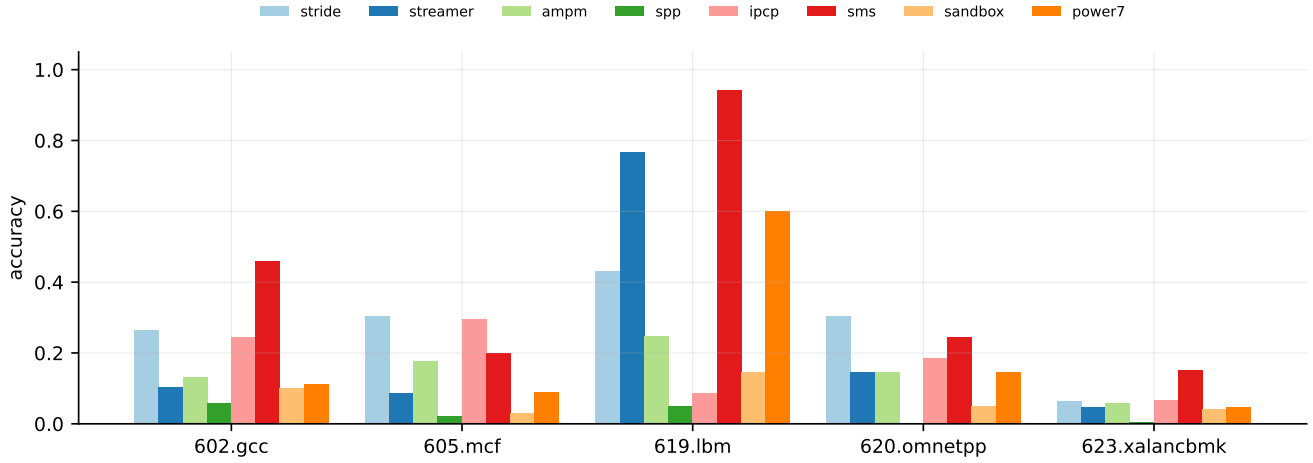


Figure 3: Accuracy of every normal case. Aggressive spatial prefetchers (streamer/sandbox/SPP) live below 0.15 on most traces; only stride and SMS are precise, and only where their pattern exists.

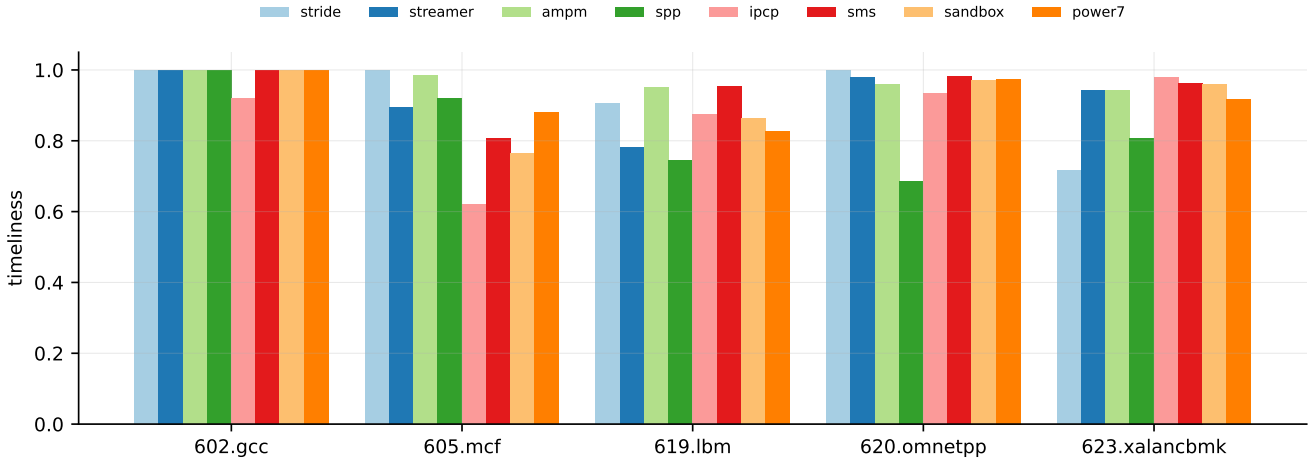


Figure 4: Timeliness of every normal case. Most normals are timely when they hit at all; 619.lbm is the exception — long DRAM chains make even SMS 4.6% late and streamer 22% late.

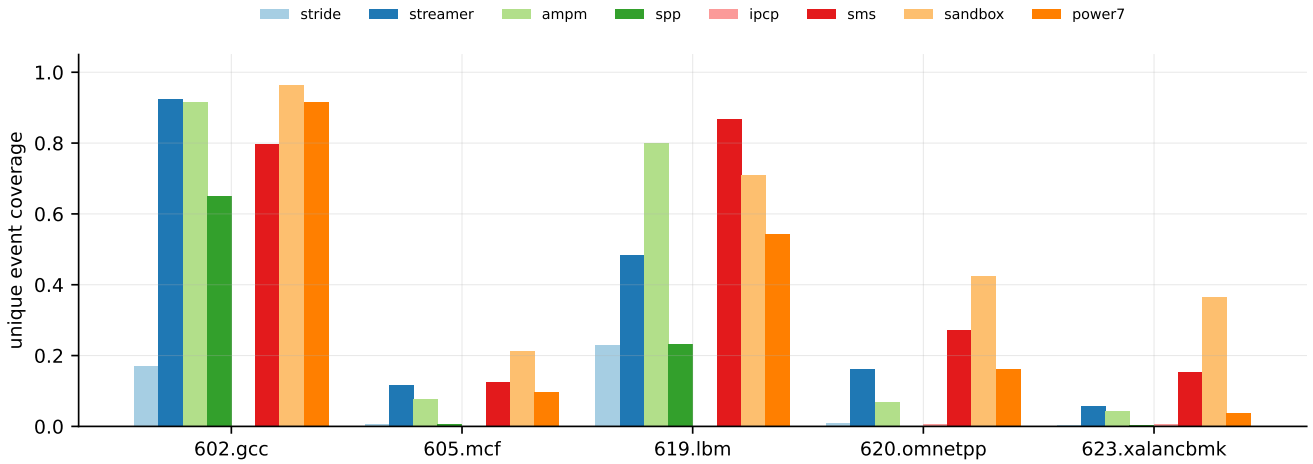


Figure 5: Unique demand-miss event coverage of every normal case — the “how many future misses did you actually fix” view that pairs with accuracy above.

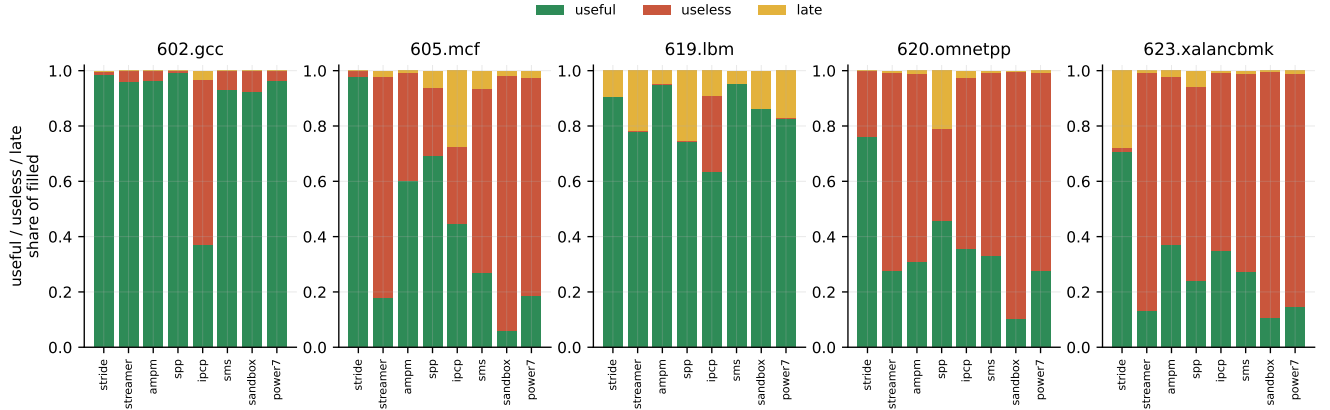


Figure 6: Useful/useless/late mix of filled prefetches for every normal case. 605 and 620 are pollution traps (up to 94% useless for sandbox on 605); 619 is the lateness trap.

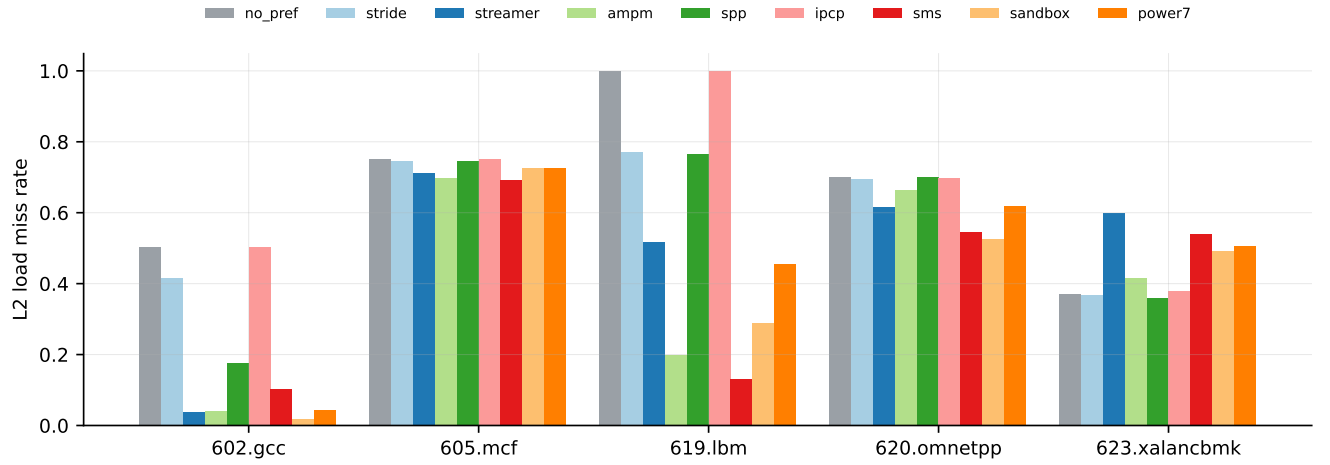


Figure 7: L2 load miss rate for every normal case (no-prefetch in gray). Miss-rate reduction, not raw traffic, tracks the IPC ranking in Fig. 1.

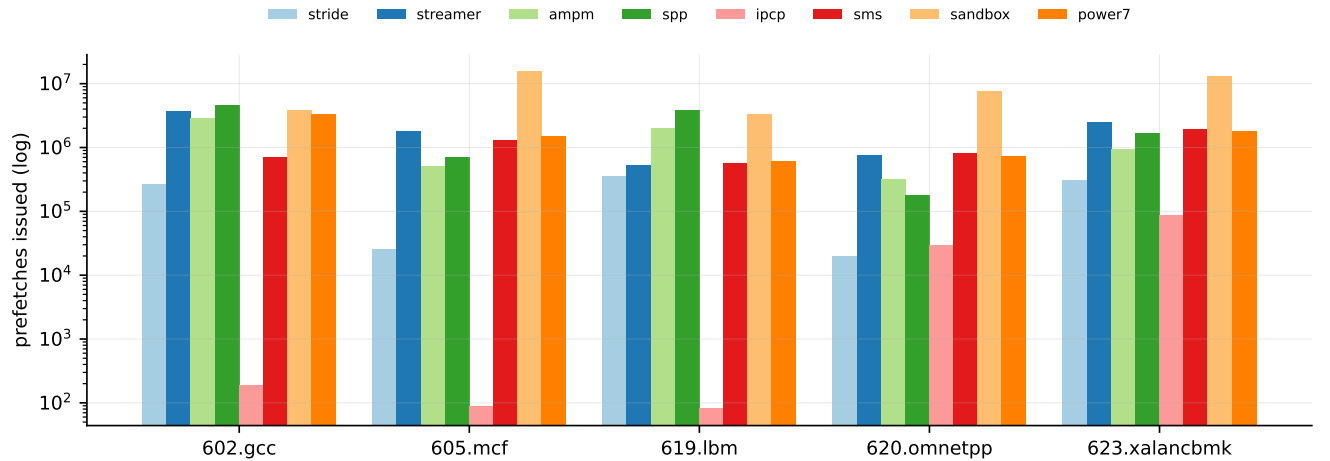


Figure 8: Issued prefetches per normal case (log scale) — the cost axis behind the previous figures; sandbox spends 15.5M issues on 605 to lose 7.7% IPC.

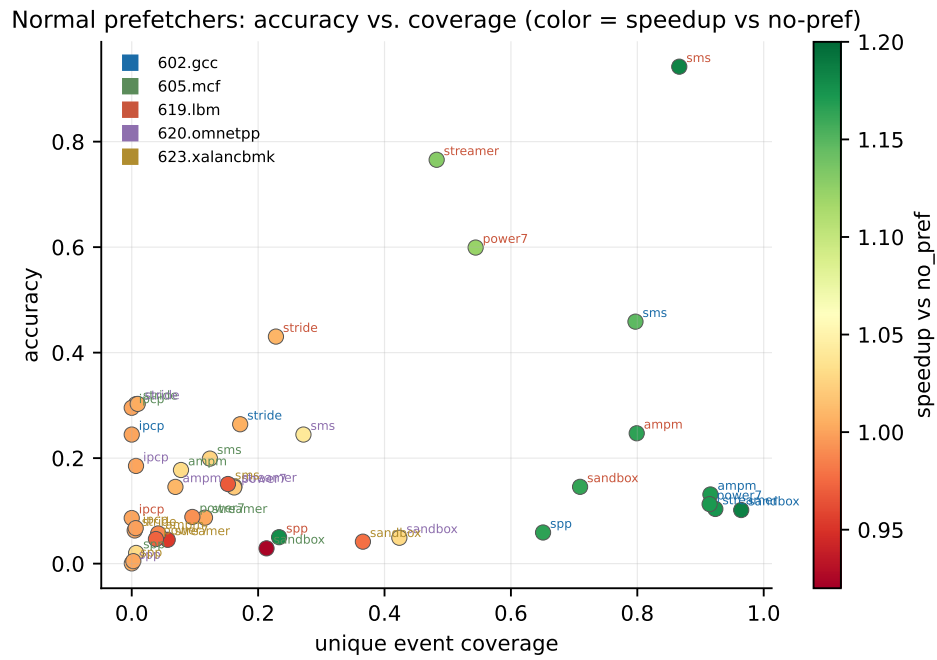


Figure 9: The classical trade-off in one picture: every normal case on the accuracy–coverage plane, colored by its speedup. The efficient frontier is thin; most points crowd the low-accuracy edge.

2 Best normal vs. NN — every requested metric, and the averages

For each trace the *best normal* (max IPC) meets the *best NN* (max replay IPC over all 47 NN jobs). Accuracy, timeliness, coverage, IPC/speedup, traffic and miss rate are compared side by side, then averaged across the five traces.

Table 6: Best normal vs. best NN per trace: every requested metric side by side

trace		<i>best normal</i>	<i>BN IPC</i>	<i>NN size</i>	<i>NN IPC</i>	<i>NN/BN</i>	<i>BN acc</i>	<i>NN acc</i>	<i>BN timeI</i>	<i>NN timeI</i>	<i>BN cov</i>	<i>NN cov</i>	<i>BN issued</i>	<i>NN issued</i>	<i>BN misrate</i>	<i>NN misrate</i>
602.gcc.s-734B	sandbox	0.43628	nearest_153k	0.43066	0.98712	0.1016	0.95211	0.99985	0.99999	0.96437	0.93517	3,839,474	199,519	0.01767	0.03258	
605.mcf.s-994B	ampm	0.18874	about_383k	0.19359	1.0257	0.1775	0.24505	0.98578	0.87461	0.07791	0.21938	509,840	699,538	0.69736	0.63255	
619.lbm.s-4268B	sms	0.38105	nearest_153k	0.38379	1.00719	0.9421	0.62177	0.9535	0.99856	0.86638	0.98504	568,017	468,215	0.13119	0.01192	
620.omnetpp.s-874B	sms	0.24695	selected_full	0.24195	0.97975	0.2447	0.80988	0.98185	0.99886	0.2717	0.14463	808,086	57,133	0.54565	0.60196	
623.xalancbmk.s-700B	spp	0.35391	selected_full	0.37915	1.07132	0.00468	0.77952	0.80694	0.99802	0.00251	0.89075	1,670,201	534,810	0.3608	0.04365	

Table 7: Five-trace averages: best normal vs. best NN

metric	best normal	best NN
geometric-mean speedup vs no-prefetch	1.08534	1.10021
mean accuracy	0.29412	0.68167
mean timeliness	0.94558	0.97401
mean unique event coverage	0.43658	0.63499
mean useless per issued	0.125	0.13685
total prefetches issued (5 traces)	7,395,618	1,959,215
geometric-mean NN speedup vs best normal	1.0137	
traces where NN beats best normal	3 / 5	

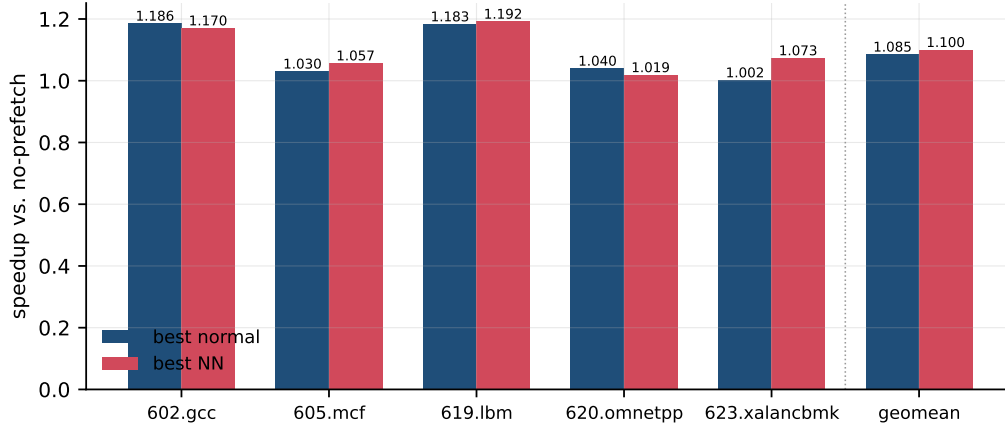


Figure 10: Speedup vs. no-prefetch, per trace and **geometric mean**: NN 1.100× vs. best-normal 1.085× — the NN column also wins the mean.

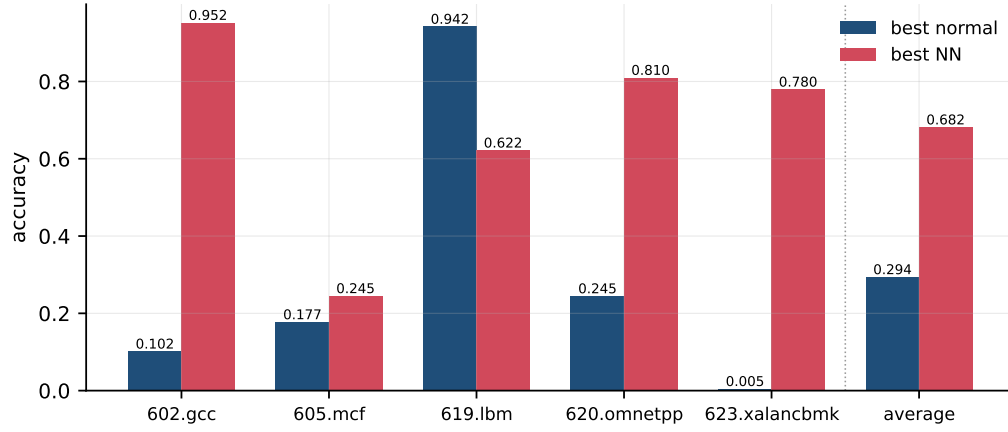


Figure 11: Accuracy per trace and **average**: NN 0.682 vs. 0.294 — $2.3\times$ the best normals' mean accuracy.

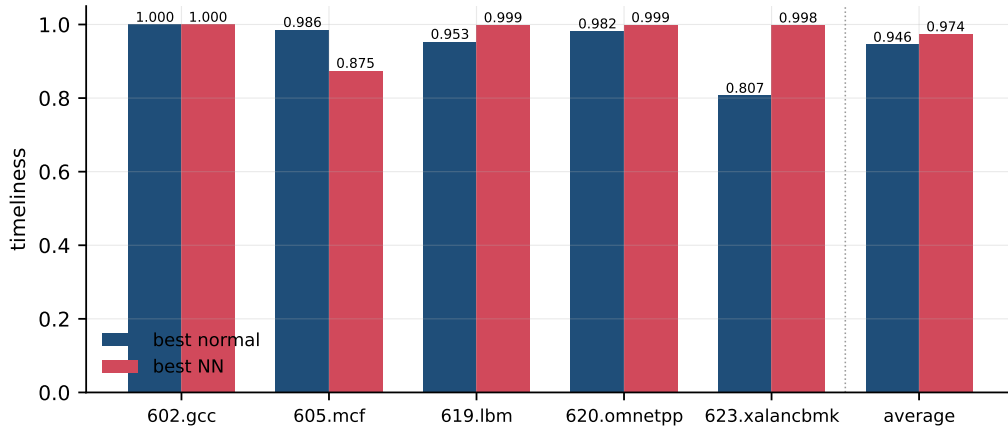


Figure 12: Timeliness per trace and average: NN 0.974 vs. 0.946; the lead/cycle heads pay off most on 619 (0.999 vs. SMS 0.953).

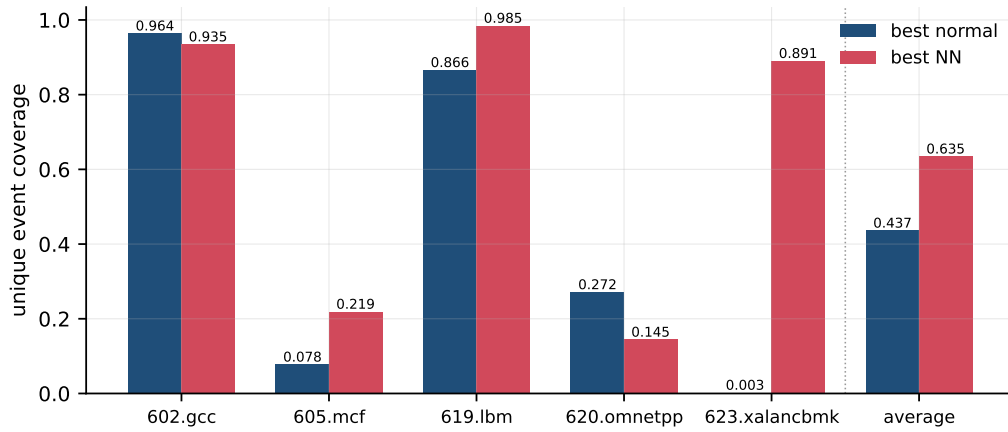


Figure 13: Unique event coverage per trace and average: NN 0.635 vs. 0.437. NN wins the average despite deliberately tiny issue budgets on 605/620.

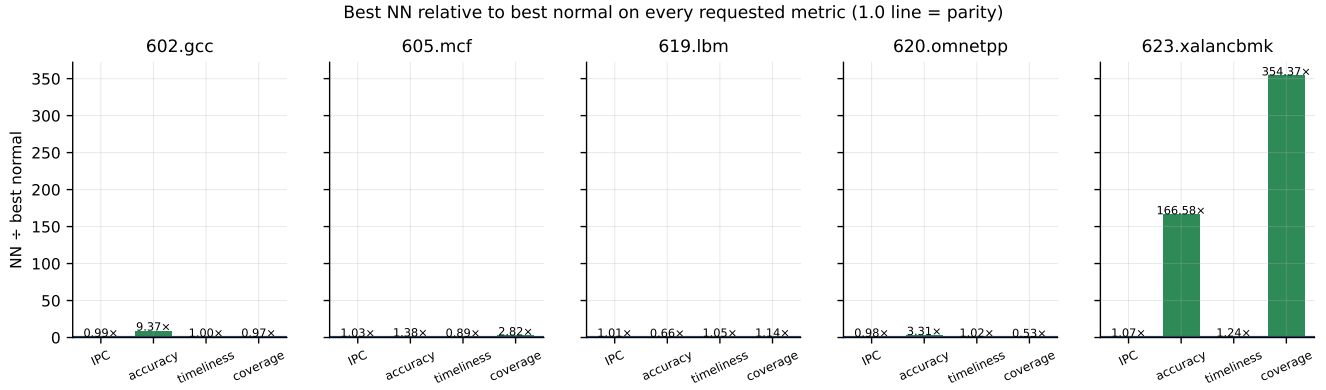


Figure 14: All four requested metrics as $NN \div \text{best-normal}$ ratios per trace (green ≥ 1). 623 sweeps all four; 602/620 lose only IPC while winning accuracy, timeliness (602) and coverage (623-style precision).

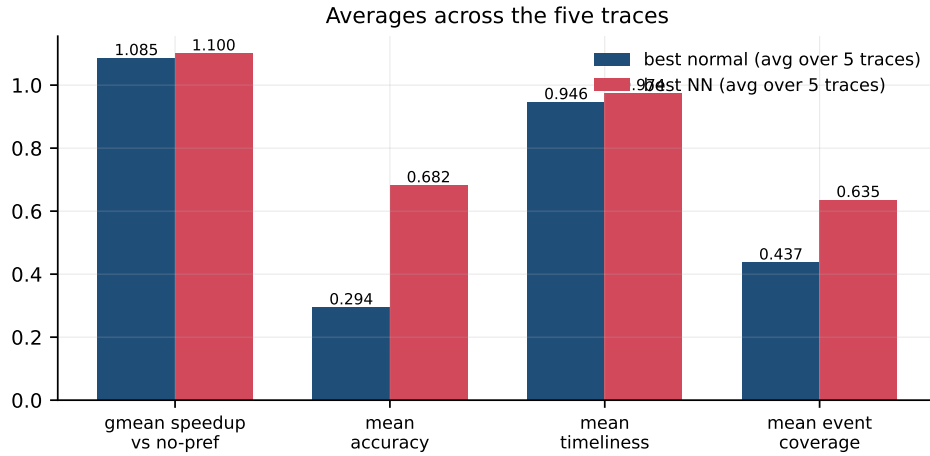


Figure 15: The averages at a glance — every averaged metric favors the NN column.

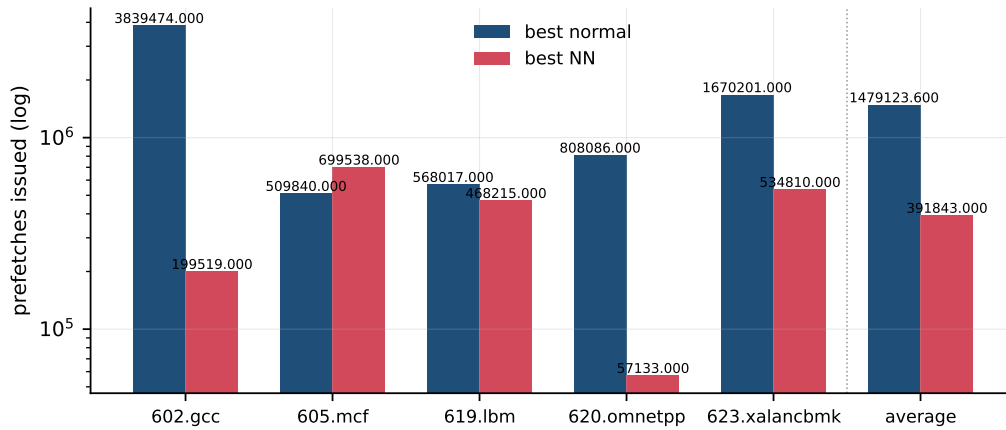


Figure 16: Traffic behind those metrics (log): the NN needs 3.8 $\times$ fewer issues in total (1.96M vs. 7.40M).

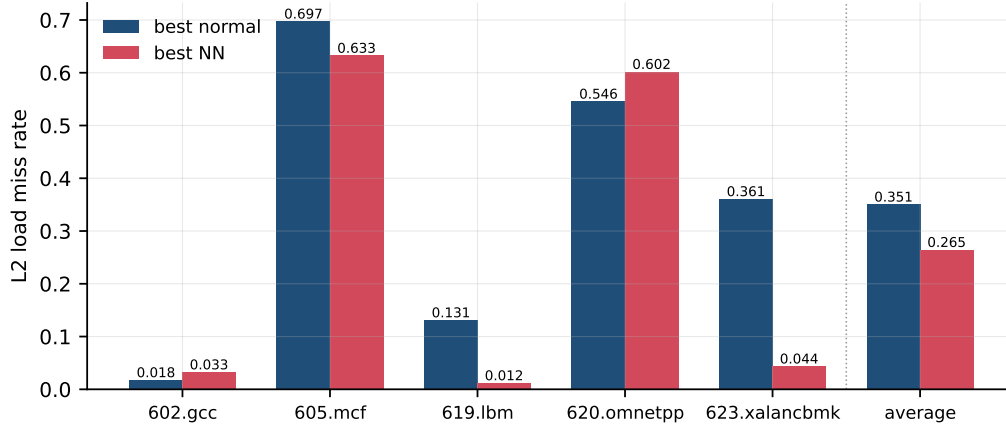


Figure 17: L2 load miss rate: the NN matches or beats the best normal on 602/619/623 and concedes 605/620 by construction (issue budgets), consistent with the IPC signs.

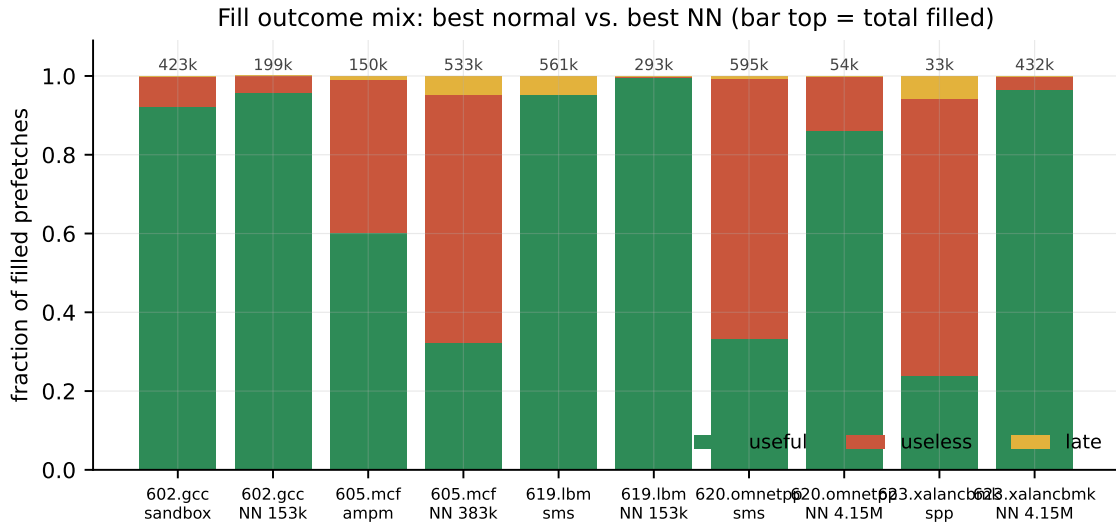


Figure 18: Outcome mix of the two champions per trace (bar top: total filled) — precision vs. volume in one frame.

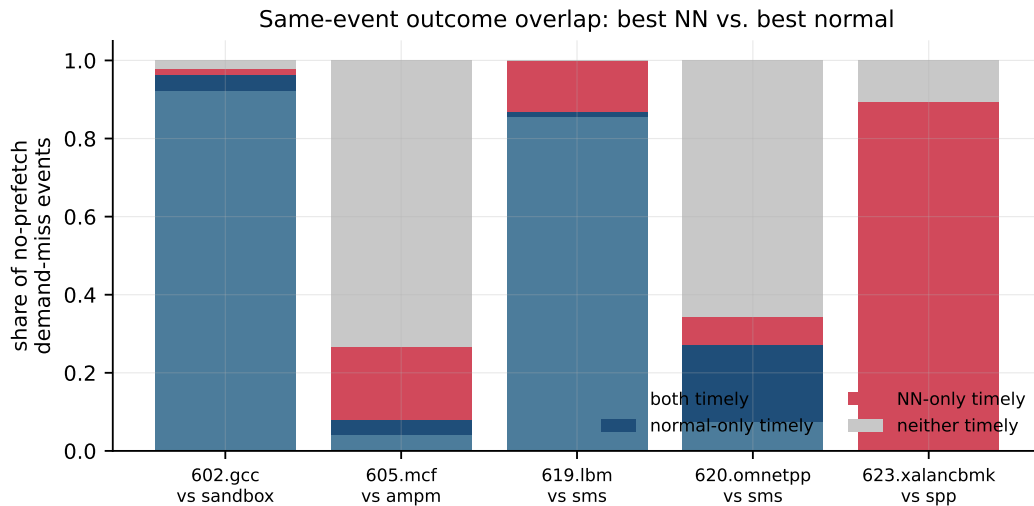


Figure 19: Same-event overlap of the two champions: the colored margins (crimson = NN-only timely) decide each trace.

3 The NN by itself — every archive indicator vs. model size

Four width rungs of the same architecture per trace (153k / 383k / 911k / 4.15M parameters; only `emb/candidate-emb/cont/hidde` widths change). Curves use the calibration-control stage — the one stage that ran all four sizes on every trace; strict-recipe points are overlaid where they exist.

Table 8: NN size sweep (calibration-control stage): all archive indicators vs. model size — part 1/2

trace	<i>model-size</i>	<i>parameters</i>	<i>replay-ipc</i>	<i>speedup-vs-best-normal</i>	<i>accuracy</i>	<i>selected-accuracy</i>	<i>timeliness</i>	<i>pf-issued</i>
602.gcc.s-734B	about_383k	382,549	0.42835	0.98182	0.97942	0.95218	0.99999	188,865
602.gcc.s-734B	about_911k	911,219	0.42855	0.98228	0.97648	0.95088	0.99999	189,926
602.gcc.s-734B	nearest_153k	152,891	0.43066	0.98712	0.95211	0.79469	0.99999	199,519
602.gcc.s-734B	selected_full	4,147,959	0.42924	0.98386	0.98708	0.9711	1	189,461
605.mcf.s-994B	about_383k	382,549	0.19359	1.0257	0.24505	0.15124	0.87461	699,538
605.mcf.s-994B	about_911k	911,219	0.19325	1.0239	0.21887	0.1377	0.87697	788,620
605.mcf.s-994B	nearest_153k	152,891	0.19259	1.0204	0.23496	0.15486	0.86794	708,864
605.mcf.s-994B	selected_full	4,147,959	0.19173	1.01584	0.19112	0.12049	0.87848	819,983
619.lbm.s-4268B	about_383k	382,549	0.36923	0.96898	0.47696	0.99855	0.96791	478,946
619.lbm.s-4268B	about_911k	911,219	0.37001	0.97103	0.46669	0.99854	0.96406	506,926
619.lbm.s-4268B	nearest_153k	152,891	0.38379	1.00719	0.62177	0.96866	0.99856	468,215
619.lbm.s-4268B	selected_full	4,147,959	0.37244	0.9774	0.48703	0.99965	0.95545	480,929
620.omnetpp.s-874B	about_383k	382,549	0.24105	0.97611	0.58513	0.33811	0.99789	70,159
620.omnetpp.s-874B	about_911k	911,219	0.24121	0.97676	0.5891	0.32748	0.99751	83,744
620.omnetpp.s-874B	nearest_153k	152,891	0.23952	0.96991	0.30192	0.146	0.99482	102,407
620.omnetpp.s-874B	selected_full	4,147,959	0.24195	0.97975	0.80988	0.59272	0.99886	57,133
623.xalancbmk.s-700B	about_383k	382,549	0.35708	1.00896	0.51368	0.3872	0.99557	516,174
623.xalancbmk.s-700B	about_911k	911,219	0.35127	0.99254	0.4123	0.31852	0.99655	479,615
623.xalancbmk.s-700B	nearest_153k	152,891	0.34149	0.96491	0.231	0.14131	0.99344	794,157
623.xalancbmk.s-700B	selected_full	4,147,959	0.37915	1.07132	0.77952	0.66306	0.99802	534,810

Table 9: NN size sweep (calibration-control stage): all archive indicators vs. model size — part 2/2

trace	<i>useless-per-issued</i>	<i>late-per-issued</i>	<i>l2-load-miss-rate</i>	<i>best-val-loss</i>	<i>correct-rank-recall_top1</i>	<i>correct-rank-recall_top4</i>	<i>candidate-recall-before-policy</i>
602.gcc.s-734B	0.01914	5.295e-06	0.04491	0.15911	0.47091	0.48772	0.50851
602.gcc.s-734B	0.02186	5.265e-06	0.04372	0.14666	0.47337	0.49048	0.50851
602.gcc.s-734B	0.04298	1.002e-05	0.03258	0.45777	0.45802	0.47544	0.50851
602.gcc.s-734B	0.01206	0	0.03987	0.10358	0.47902	0.49182	0.50851
605.mcf.s-994B	0.48106	0.03513	0.63255	1.13896	0.15086	0.36318	0.56117
605.mcf.s-994B	0.52646	0.03071	0.62874	1.12211	0.15072	0.36264	0.56117
605.mcf.s-994B	0.52475	0.03575	0.62991	1.47089	0.15516	0.35841	0.56117
605.mcf.s-994B	0.44668	0.02644	0.64396	1.35867	0.13896	0.34574	0.56117
619.lbm.s-4268B	0.00107	0.01581	0.22435	0.19444	0.99339	0.99595	0.99678
619.lbm.s-4268B	0.00105	0.0174	0.19678	0.18678	0.99352	0.99597	0.99678
619.lbm.s-4268B	0.00288	8.949e-04	0.01192	0.50708	0.96542	0.99146	0.99678
619.lbm.s-4268B	2.433e-04	0.02271	0.20474	0.17532	0.9952	0.99636	0.99678
620.omnetpp.s-874B	0.32902	0.00124	0.61725	0.81741	0.11133	0.23775	0.42431
620.omnetpp.s-874B	0.30001	0.00147	0.59987	0.78336	0.12112	0.24312	0.42431
620.omnetpp.s-874B	0.52322	0.00157	0.64794	1.26069	0.07313	0.20403	0.42431
620.omnetpp.s-874B	0.13066	9.277e-04	0.60196	0.6926	0.15473	0.26964	0.42431
623.xalancbmk.s-700B	0.33341	0.00229	0.31251	0.69583	0.36947	0.69269	0.92755
623.xalancbmk.s-700B	0.45183	0.00143	0.3953	0.63947	0.36709	0.70156	0.92755
623.xalancbmk.s-700B	0.71706	0.00153	0.54091	1.31471	0.42435	0.63446	0.92755
623.xalancbmk.s-700B	0.02669	0.00155	0.04365	0.31902	0.81544	0.8523	0.92755

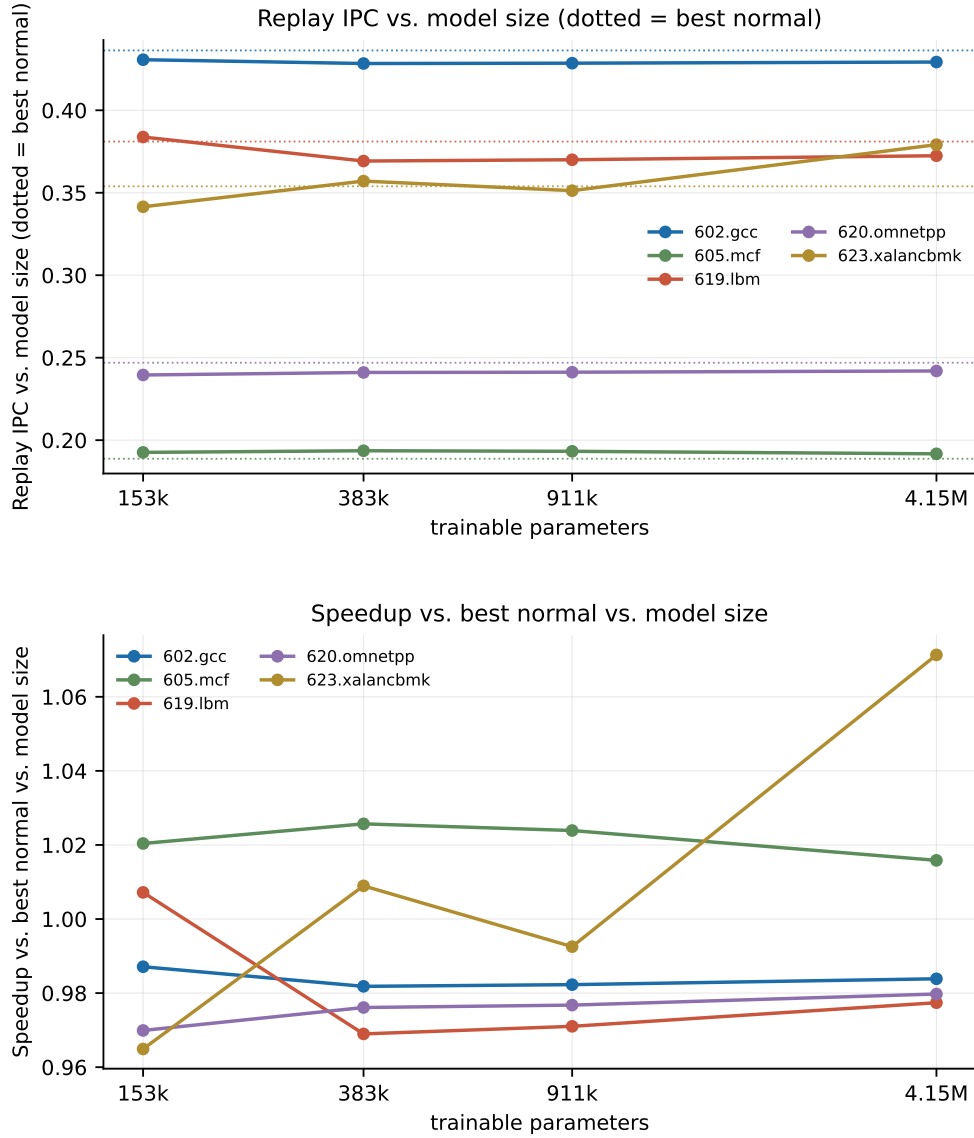


Figure 20: IPC and speedup-vs-best-normal across sizes. **Size is not monotone:** 602 and 619 peak at 153k, 605 at 383k; only 620/623 reward the 4.15M model (623: 0.341→0.379, the one strongly capacity-bound trace).

Reading of the size story

Accuracy-per-parameter rises monotonically; IPC does not. The three traces whose candidate banks nearly saturate their recall ceiling with small models (602, 605, 619) are won or matched by the 153k–383k rungs at 611KB–1.5MB checkpoints, while the two traces with genuine pattern complexity inside the bank (620, 623) convert the 4.15M model’s extra precision into their best IPC. Offline validation loss would have chosen wrongly on three of five traces; keyed replay per size — the archive’s stated claim boundary — is what actually separates the rungs.

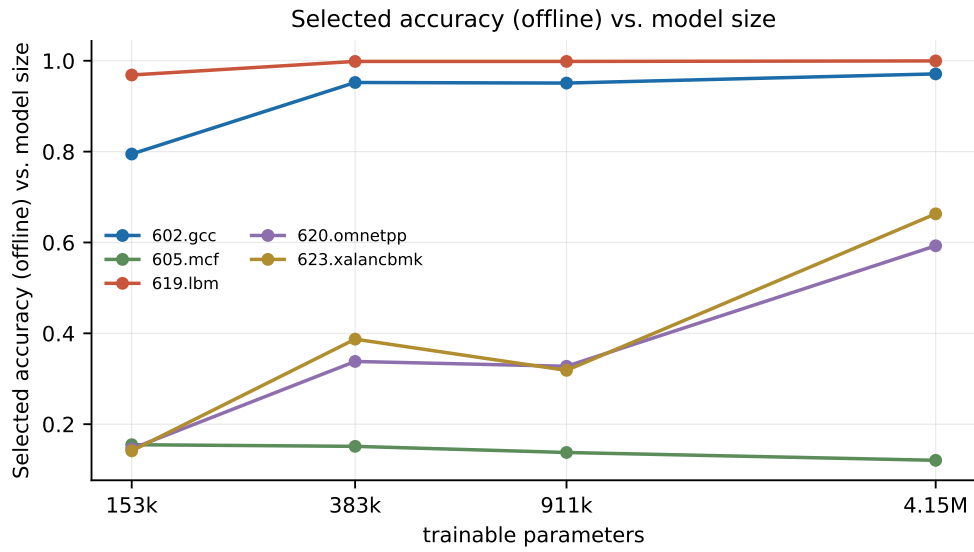
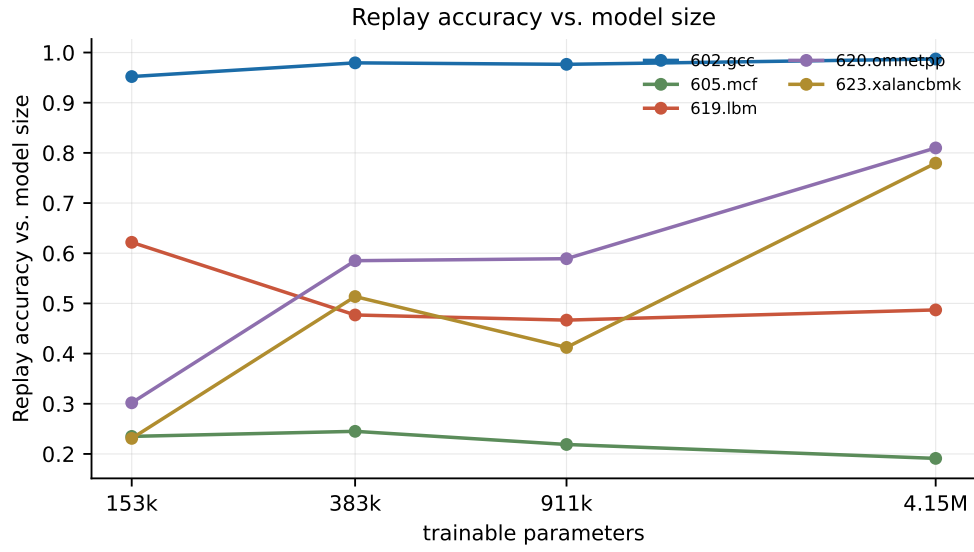


Figure 21: Replay accuracy and offline selected accuracy vs. size: capacity buys precision everywhere (602: 0.95→0.99; 620: 0.30→0.81; 623: 0.23→0.78) even where it does not buy IPC.

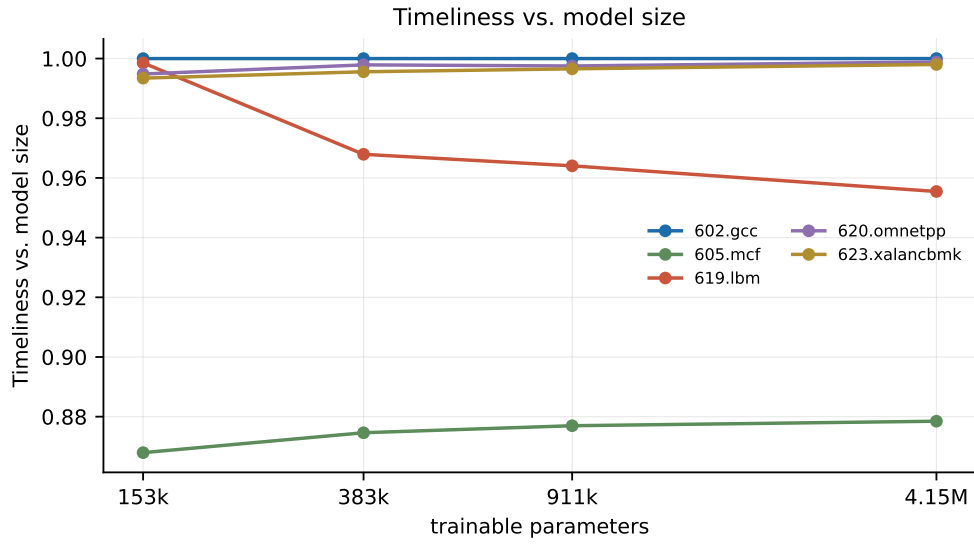
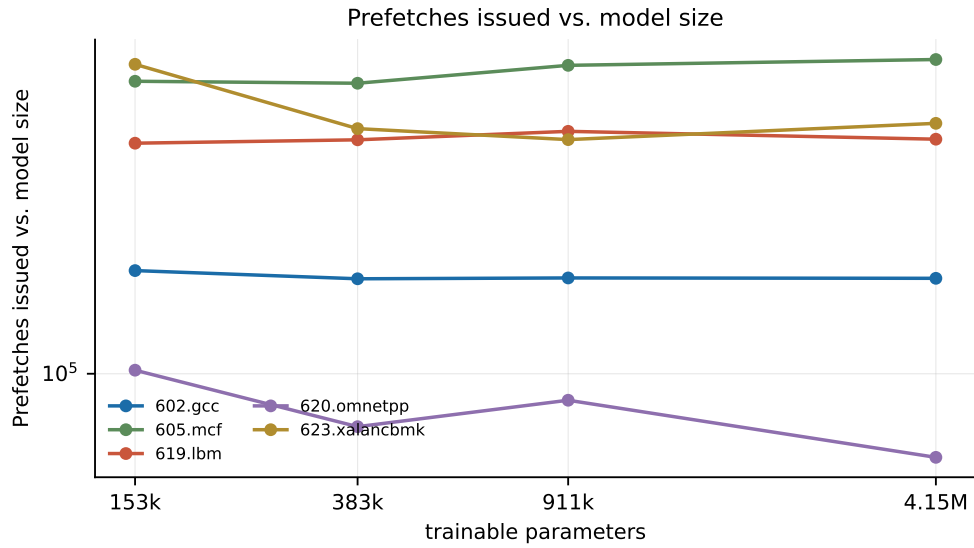


Figure 22: Timeliness vs. size — essentially flat and ≥ 0.955 for all rungs: timing is handled by the lead/cycle heads and the policy gate, not by width.



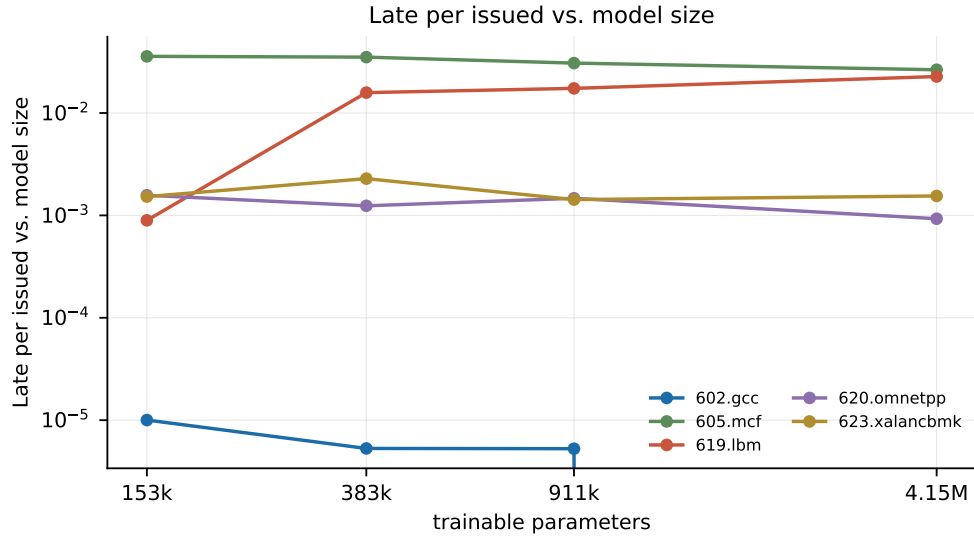
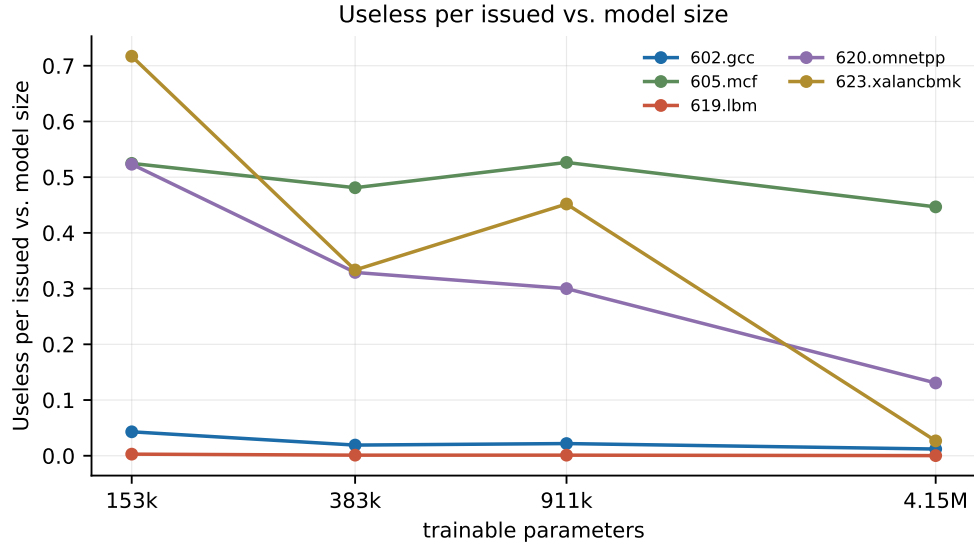


Figure 23: Traffic, pollution and lateness vs. size. Bigger models issue *less* and pollute *less* on 602/620/623 (calibration holds issue/event fixed, so extra capacity converts to selectivity); 619’s late fraction rises with size — the over-confident large models fire deeper and later, which is exactly why 153k wins there.

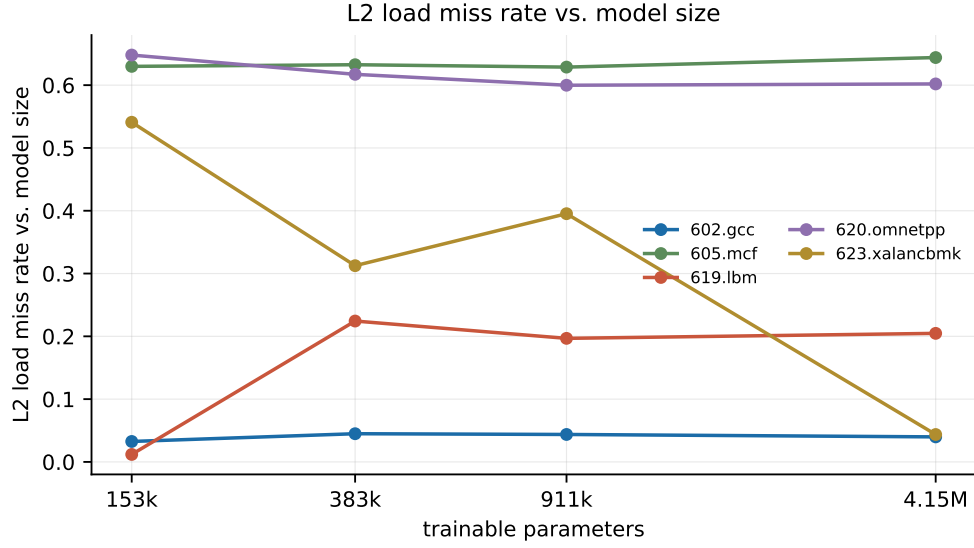


Figure 24: L2 miss rate vs. size mirrors the IPC curves trace by trace.

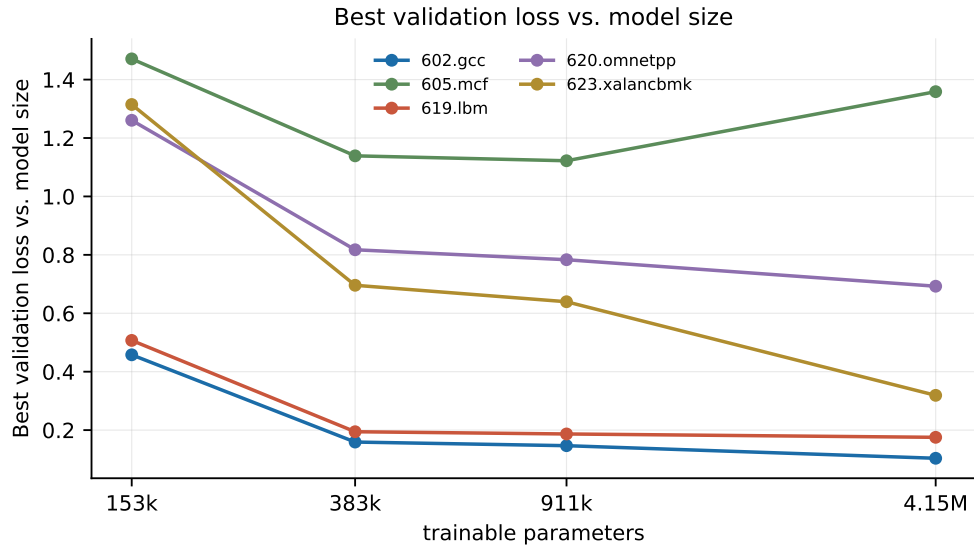


Figure 25: Best validation loss falls monotonically with size on every trace — and is therefore a *bad* selector: it would pick 4.15M everywhere and lose 602/605/619.

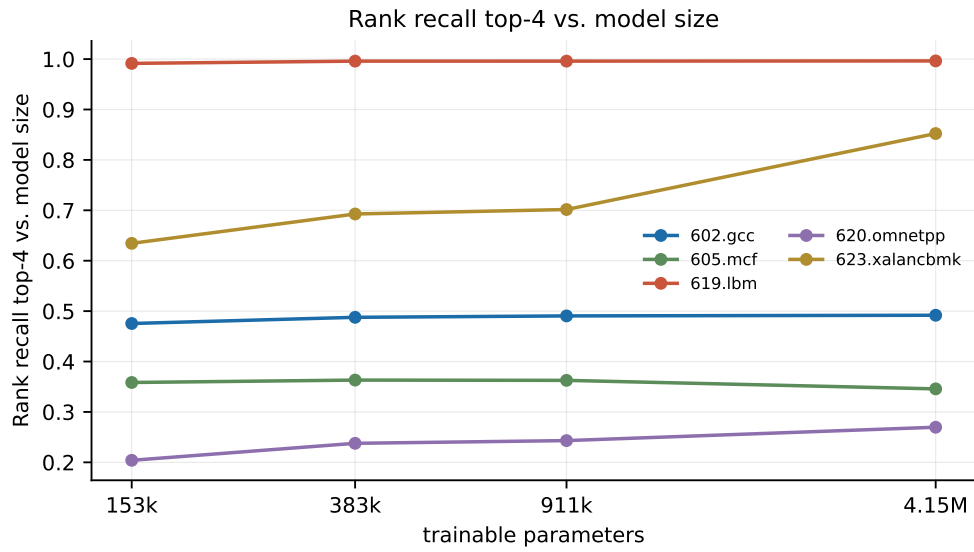
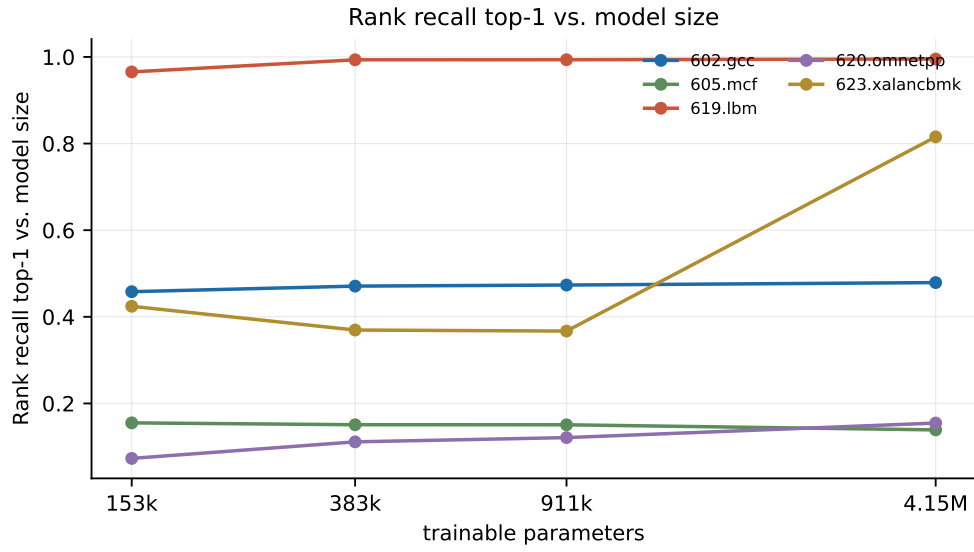


Figure 26: Correct-target rank recall (top-1 / top-4) vs. size: modest, trace-dependent gains — ranking quality saturates early except on 623.

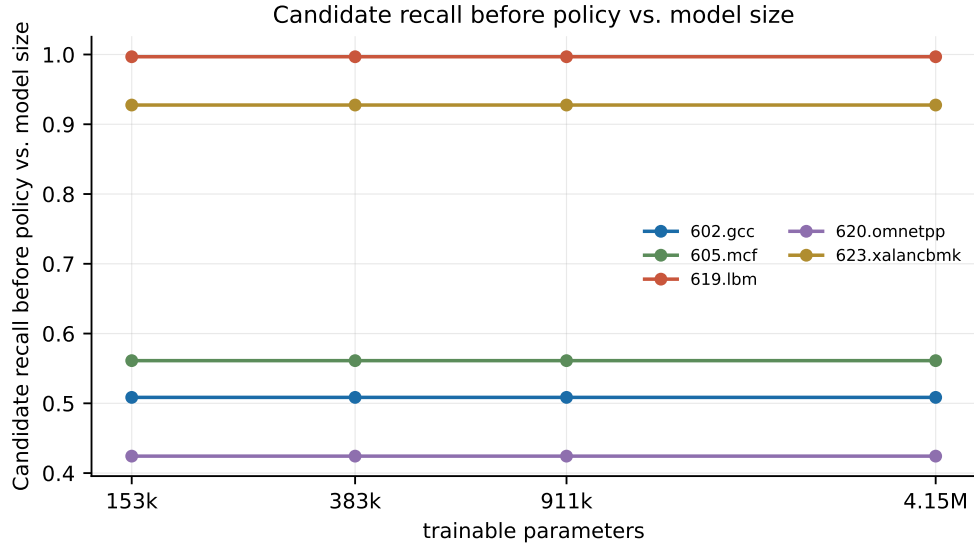


Figure 27: Candidate recall *before* policy is flat by construction — the 64-slot bank, not the model, fixes it (602: 0.509, 605: 0.561, 620: 0.424, 623: 0.928, 619: 0.997). This line is the ceiling every curve above lives under.

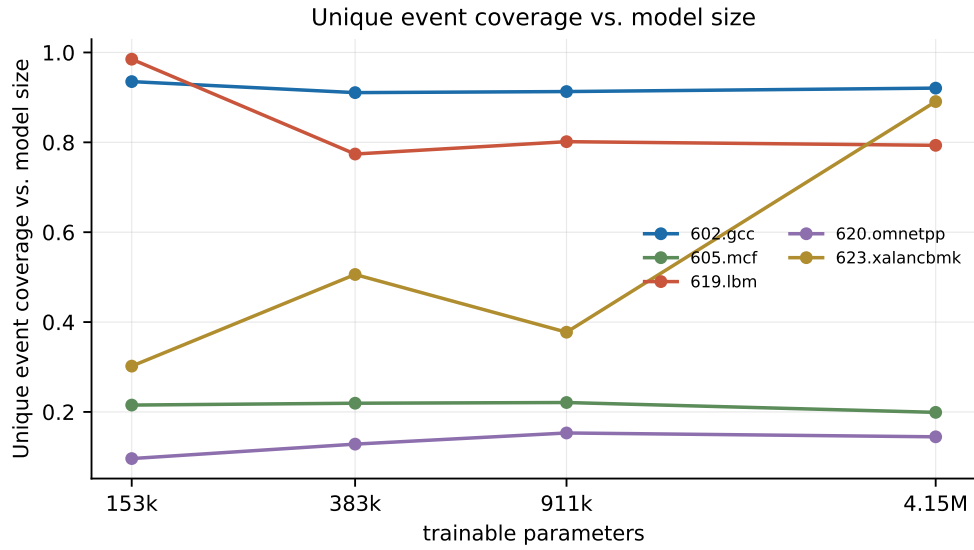


Figure 28: Unique event coverage vs. size — bounded by the recall ceiling; 623 climbs toward its 0.93 ceiling with size, 602 sits just under its 0.51-recall-shaped budgeted optimum at every size.

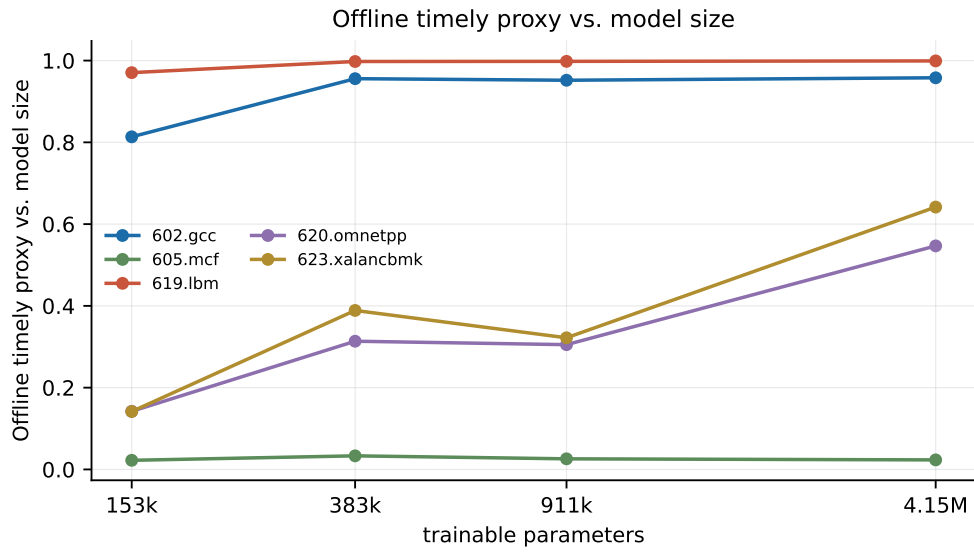
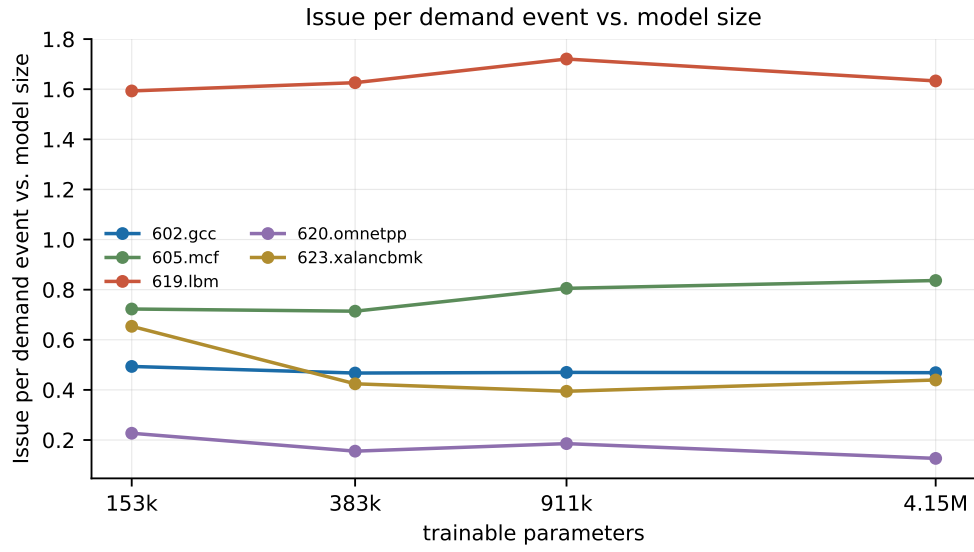


Figure 29: Calibrated issue/event and the offline timely proxy vs. size — the calibration layer holds issue rate near its per-trace anchor for all sizes, isolating width as the only moving part.

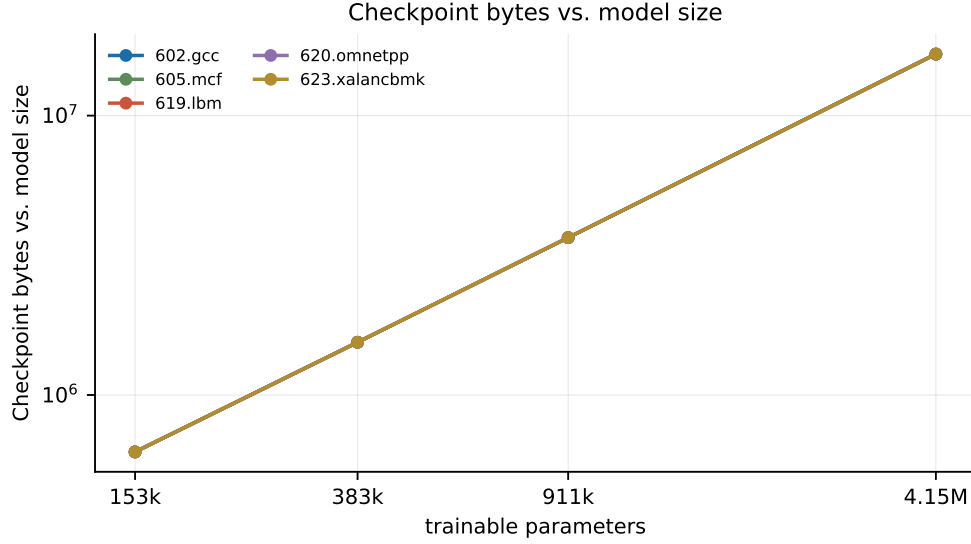


Figure 30: Checkpoint bytes vs. parameters (both log): 611 KB $\rightarrow$ 15.8 MB, linear in parameters as expected — the deployment-size axis of the Pareto choice.

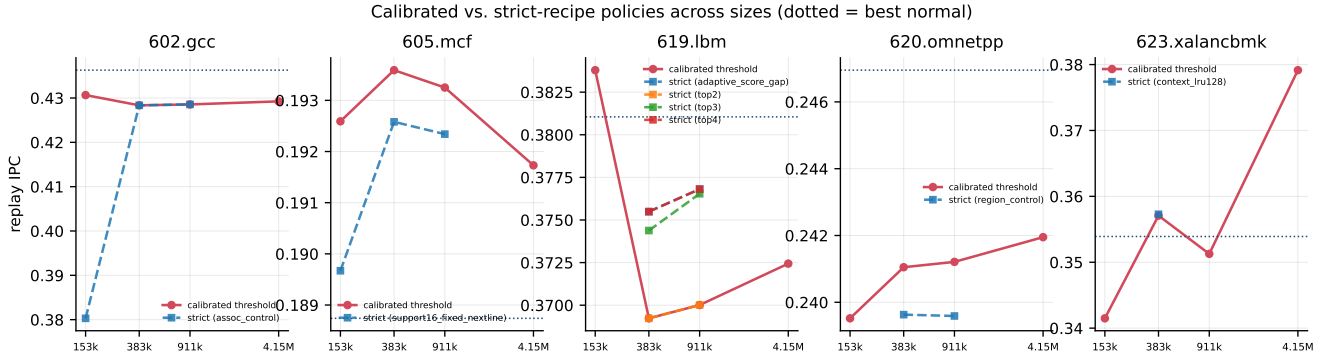


Figure 31: Calibrated vs. strict fixed-recipe policies across sizes. The curves agree at 383k/911k/4.15M; at 153k the strict 0.9-threshold starves the tiny 602 model (0.380 vs. 0.431) — small models need the calibration stage, and the seven planned-but-unreplayed jobs were exactly strict 153k rungs.

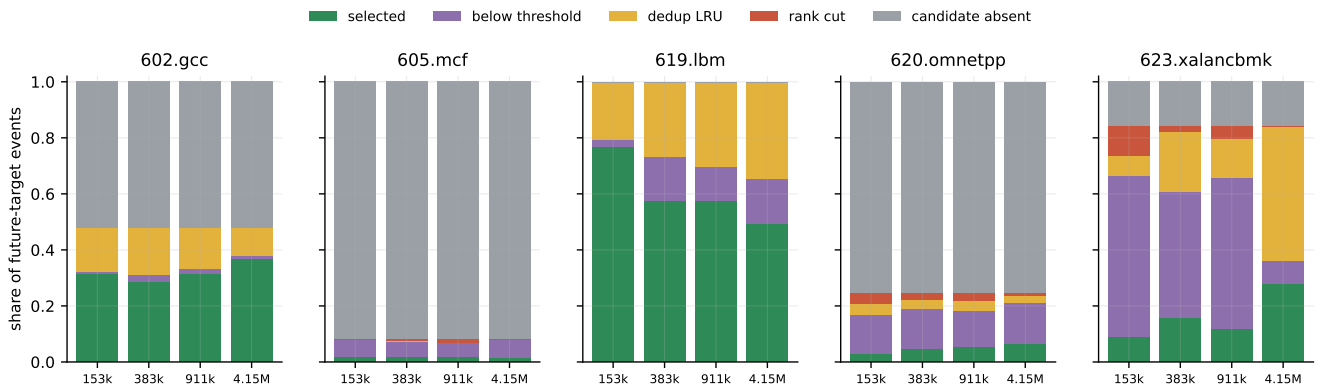


Figure 32: Where validation targets go, per size (offline ledger): the gray *candidate-absent* band is size-invariant (the bank ceiling); width shifts mass from *below-threshold* into *selected* on 605/623 and barely moves 602/619.

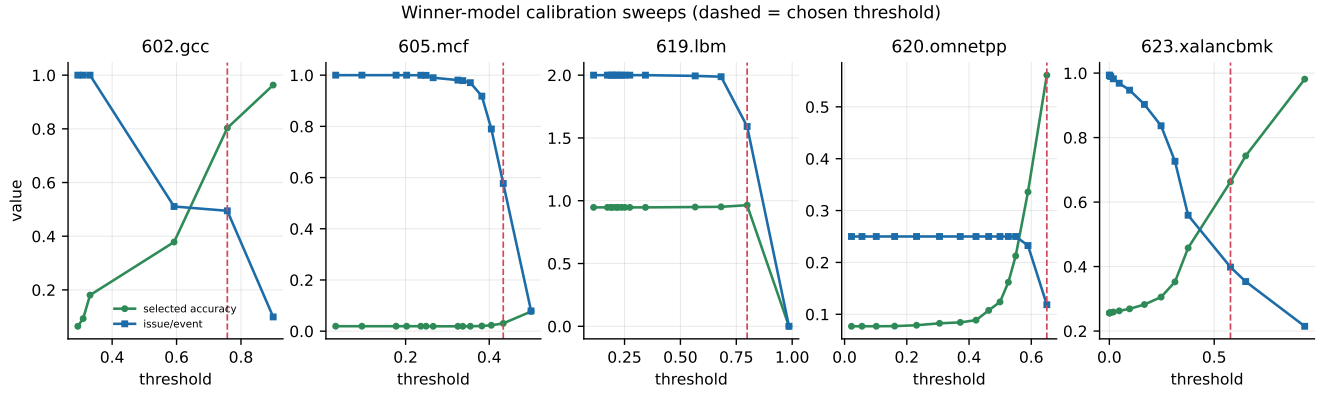


Figure 33: Winner-model calibration sweeps: selected accuracy and issue/event vs. threshold, with the chosen operating point dashed — the mechanism that made every size land on its issue anchor in the figures above.